THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Number Play** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 1234	End with the smallest even digit, then arrange the rest in increasing order.
2	**(c)** 90	First digit 1-9 and middle digit 0-9, the last digit is fixed.
3	**(c)** 29992	Fill the middle digits with 9.
4	**(a)** 17	Count each step.
5	**(c)** 376	Subtract.
6	**(a)** 465	Triangular number.
7	**(b)** 6174	Largest - smallest.
8	**(d)** 9100	Put 9s from the left until the sum is used up.
9	**(c)** 2576	End with the smallest even digit, then arrange the rest in increasing order.
10	**(c)** 49994	Fill the middle digits with 9.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Greatest 97431 (1); Smallest 13479 (1).	1+1
12	87, 165, 726, 1353, 4884 (1); Palindrome 4884 (1).	1+1
13	Estimate $= 8800 - 3200 = 5600$ (1); Exact $= 5566$ (1).	1+1
14	Greatest 86520 (1); Smallest 20568 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Step 1: 8172 (1); Step 2: 7443 (1); Digits used: arrange 8172 to get the next number (1).	1+1+1
16	Largest below: 1221 (1); Smallest above: 1331 (1); Check by reversing (1).	1+1+1
17	Pairs sum to 41 (1); Number of pairs $= 40 \div 2 = 20$ (1); Sum $= 20 \times 41 = 820$ (1).	1+1+1
	OR: Step 1: 6264 (1); Step 2: 4176 (1); Digits used: arrange 6264 to get the next number (1).	1+1+1
18	Largest below: 2442 (1); Smallest above: 2552 (1); Check by reversing (1).	1+1+1
	OR: Pairs sum to 61 (1); Number of pairs $= 60 \div 2 = 30$ (1); Sum $= 30 \times 61 = 1830$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Middle digits equal; $2 \times 2 + 2m = 10$ so $m = 3$ (1); Number 2332 (1); Reverse is the same: palindrome (1); Add 11: 2343 (1); Not a palindrome (1).	1+1+1+1+1
	OR: Try arrangements (1); Closest number is 5137 (2); Difference $= 137$ (2).	1+2+2
20	Middle digits equal; $2 \times 3 + 2m = 14$ so $m = 4$ (1); Number 3443 (1); Reverse is the same: palindrome (1); Add 11: 3454 (1); Not a palindrome (1).	1+1+1+1+1
	OR: Try arrangements (1); Closest number is 6248 (2); Difference $= 248$ (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Number Play** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 7	Count each step.
2	**(a)** 250	Subtract.
3	**(c)** 1035	Triangular number.
4	**(b)** 8532	Largest - smallest.
5	**(c)** 9600	Put 9s from the left until the sum is used up.
6	**(b)** 1398	End with the smallest even digit, then arrange the rest in increasing order.
7	**(d)** 69996	Fill the middle digits with 9.
8	**(c)** 10	Count each step.
9	**(b)** 52	Subtract.
10	**(c)** 1830	Triangular number.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	96, 165, 726, 1353, 4884 (1); Palindrome 4884 (1).	1+1
12	Estimate $= 9300 - 4700 = 4600$ (1); Exact $= 4647$ (1).	1+1
13	Greatest 97531 (1); Smallest 13579 (1).	1+1
14	175, 746, 1393, 5324, 9559 (1); Palindrome 9559 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Step 1: 6174 (1); Step 2: 6174 (1); Digits used: arrange 6174 to get the next number (1).	1+1+1
16	Largest below: 3773 (1); Smallest above: 3883 (1); Check by reversing (1).	1+1+1
17	Pairs sum to 81 (1); Number of pairs $= 80 \div 2 = 40$ (1); Sum $= 40 \times 81 = 3240$ (1).	1+1+1
	OR: Step 1: 9171 (1); Step 2: 8532 (1); Digits used: arrange 9171 to get the next number (1).	1+1+1
18	Largest below: 5005 (1); Smallest above: 5115 (1); Check by reversing (1).	1+1+1
	OR: Pairs sum to 121 (1); Number of pairs $= 120 \div 2 = 60$ (1); Sum $= 60 \times 121 = 7260$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Middle digits equal; $2 \times 4 + 2m = 10$ so $m = 1$ (1); Number 4114 (1); Reverse is the same: palindrome (1); Add 11: 4125 (1); Not a palindrome (1).	1+1+1+1+1
	OR: Try arrangements (1); Closest number is 4619 (2); Difference $= 119$ (2).	1+2+2
20	Middle digits equal; $2 \times 5 + 2m = 22$ so $m = 6$ (1); Number 5665 (1); Reverse is the same: palindrome (1); Add 11: 5676 (1); Not a palindrome (1).	1+1+1+1+1
	OR: Try arrangements (1); Closest number is 2873 (2); Difference $= 127$ (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Number Play** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 8352	Largest - smallest.
2	**(d)** 9920	Put 9s from the left until the sum is used up.
3	**(c)** 3456	End with the smallest even digit, then arrange the rest in increasing order.
4	**(a)** 89998	Fill the middle digits with 9.
5	**(a)** 111	Count each step.
6	**(d)** 420	Subtract.
7	**(a)** 5050	Triangular number.
8	**(b)** 8172	Largest - smallest.
9	**(c)** 9970	Put 9s from the left until the sum is used up.
10	**(c)** 9993	Put 9s from the left until the sum is used up.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Estimate $= 7500 - 2800 = 4700$ (1); Exact $= 4661$ (1).	1+1
12	Greatest 86420 (1); Smallest 20468 (1).	1+1
13	187, 968, 1837, 9218, 17347, 91718, 173437, 907808, 1716517, 8872688, 17735476, 85189247, 159487405, 664272356, 1317544822, 3602001953, 7193004016, 13297007933, 47267087164, 93445163438, 176881317877 (1); Palindrome 176881317877 (1).	1+1
14	Estimate $= 6500 - 2000 = 4500$ (1); Exact $= 4508$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Step 1: 6354 (1); Step 2: 3087 (1); Digits used: arrange 6354 to get the next number (1).	1+1+1
16	Largest below: 6666 (1); Smallest above: 6776 (1); Check by reversing (1).	1+1+1
17	Pairs sum to 151 (1); Number of pairs $= 150 \div 2 = 75$ (1); Sum $= 75 \times 151 = 11325$ (1).	1+1+1
	OR: Step 1: 8352 (1); Step 2: 6174 (1); Digits used: arrange 8352 to get the next number (1).	1+1+1
18	Largest below: 8118 (1); Smallest above: 8228 (1); Check by reversing (1).	1+1+1
	OR: Pairs sum to 201 (1); Number of pairs $= 200 \div 2 = 100$ (1); Sum $= 100 \times 201 = 20100$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Middle digits equal; $2 \times 6 + 2m = 16$ so $m = 2$ (1); Number 6226 (1); Reverse is the same: palindrome (1); Add 11: 6237 (1); Not a palindrome (1).	1+1+1+1+1
	OR: Try arrangements (1); Closest number is 6952 (2); Difference $= 48$ (2).	1+2+2
20	Middle digits equal; $2 \times 7 + 2m = 24$ so $m = 5$ (1); Number 7557 (1); Reverse is the same: palindrome (1); Add 11: 7568 (1); Not a palindrome (1).	1+1+1+1+1
	OR: Try arrangements (1); Closest number is 1874 (2); Difference $= 126$ (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.